SVIES: A Seven-Layer AI-Based Smart Vehicle Intelligence and Enforcement System

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***Abstract*—Existing Automatic Number Plate Recognition (ANPR) systems are restricted to plate-text extraction with no contextual vehicle intelligence or fraud-detection capability. This paper presents SVIES—the Smart Vehicle Intelligence and Enforcement System—a seven-layer (with three novel sub-layers 2.5, 4.5, and 5.5) end-to-end AI pipeline for real-world Indian road enforcement. SVIES deploys three custom-trained YOLOv8s models: plate detection at mAP@50 of 97.7% (precision 0.988), helmet detection at 74.4%, and vehicle classification at 64.4%, trained on domain-specific Indian datasets on an NVIDIA Tesla V100. A novel eight-mode Fake and Mismatched Plate Detection module performs vehicle-type mismatch analysis, CMVR color-code validation, IS 10731 font anomaly detection, duplicate plate clone alerting, state code mismatch, and cross-referencing against VAHAN, PUCC, and stolen vehicle databases. OCR uses a six-variant EasyOCR/Tesseract ensemble with Groq Llama-4 Scout 17B vision verification and position-aware correction for all Indian plate formats including BH-series. Additional contributions include a hybrid helmet/seatbelt detector, Shapely-based geofencing over 14 Hyderabad zones, three-tier offender intelligence, SHA-256 cryptographic evidence timestamping, and a YOLOv5n offline edge mode. SVIES is implemented as a containerised FastAPI + React 19 stack with real-time WebSocket streaming and Firebase RBAC.**

***Index Terms*—ANPR, YOLOv8, fake plate detection, OCR ensemble, road enforcement, India, geofencing, LLM vision, edge AI.**

I. Introduction

Road traffic enforcement in India faces a persistent technological gap. Most vehicle monitoring relies on manual policing or basic ANPR cameras limited to toll-gate plate reading [1]. Forged plates, uninsured vehicles, and habitual violators remain largely undetected [6]. The Ministry of Road Transport and Highways recorded 153,972 road fatalities in 2021, substantially attributable to helmet non-compliance, over-speeding, and unregistered vehicles [2]. India's VAHAN database covers over 300 million registered vehicles [3]; plates must conform to IS 10731 and CMVR color codes. Commercial solutions such as Genetec AutoVu target Western standards with no fraud-forensics. Singh and Mishra showed that EMI attacks degrade real-time ML camera inference [16], motivating hardened multi-layer architectures. Fig. 1 Visual Abstract gives you a full understanding architectures

SVIES transforms a live CCTV feed into actionable enforcement intelligence. Primary contributions:

- **Eight-mode Fake & Mismatched Plate Detection**: database verification (VAHAN, PUCC, insurance, stolen), CMVR color forensics, IS 10731 font metrics, and spatio-temporal clone detection.
- **Six-variant LLM-augmented OCR**: EasyOCR + Tesseract ensemble with position-aware correction and Groq Llama-4 Scout 17B vision verification covering all Indian plate formats including BH-series.
- **Hybrid helmet/seatbelt detector**: YOLOv8s cascade, YOLOv8n-pose keypoints, and Indian skin-tone HSV/YCrCb heuristic.
- **Zone-weighted risk engine**: 17 violation parameters, 4 alert tiers, Shapely geofencing over 14 Hyderabad zones with Overpass API.
- **Three-tier offender intelligence**: RTO notification $\rightarrow$ district superintendent challan $\rightarrow$ court summons.
- **SHA-256 cryptographic evidence chain** via Twilio SMS and Gmail SMTP.
- **YOLOv5n offline edge mode** with 30-day SQLite caching for Raspberry Pi 4B deployments.

II. Related Work

ANPR and OCR. Early ANPR used morphological operations and Tesseract [13]. CNN-based SSD and Faster R-CNN improved real-world detection [11], [12]. The YOLO family became the standard for real-time ANPR [4], with YOLOv8 advancing anchor-free small-object localisation [5]. Indian-specific challenges include plate degradation, mounting-angle variation, and format diversity [6]. Ensemble OCR with rule-based post-processing outperforms single-engine pipelines on regional datasets [7].

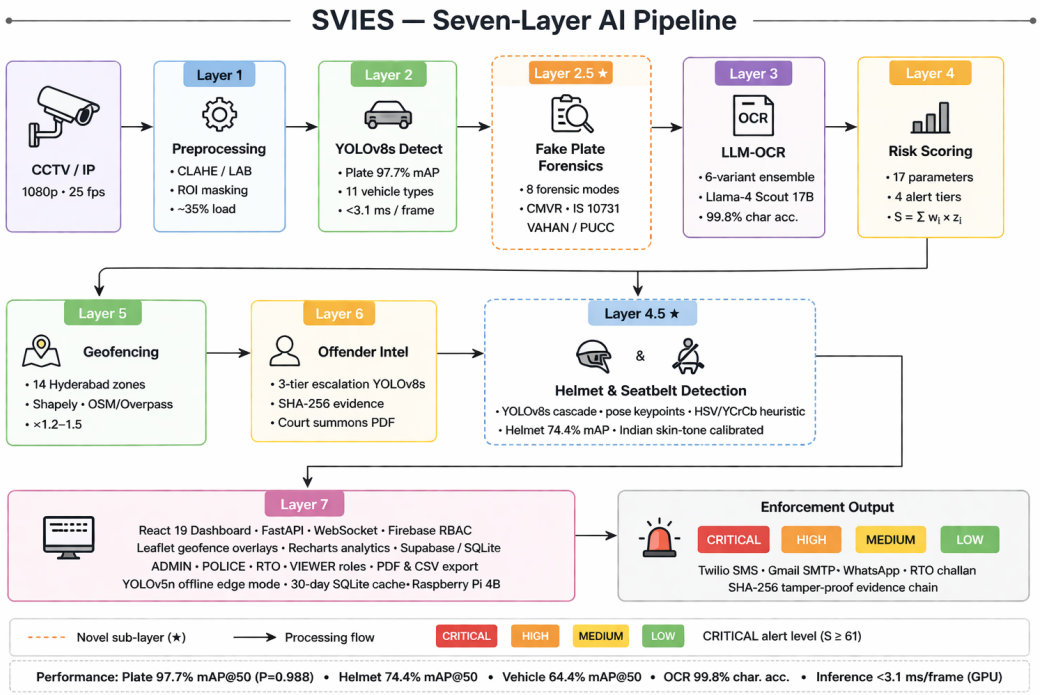


Fig. 1. Visual Abstract of SVIES: end-to-end seven-layer pipeline from live CCTV input to tiered enforcement alert.

Safety and Geofencing. YOLOv5/EfficientNet-B0 helmet detection exceeds 90% accuracy on curated data [8]. Pose-keypoint seatbelt detection and GPS-based geofencing for smart-city traffic management have been reported [9], [10].

Adversarial Hardening. Singh et al. demonstrated EMI on vehicular camera sensors reduces ML accuracy and frame rates [16]. TinyML-based IoT hardening [17] and wearable IoBT platforms [18] highlight that edge intelligence with sensor protection is essential for mission-critical enforcement [19]. Neural network vulnerability research further motivates robust multi-layer architectures [14].

III. System Architecture

SVIES is a seven-layer sequential pipeline wherein each layer processes the output of the preceding layer, contributing to a unified per-vehicle risk profile (Fig. 2).

A. Layer 1: Video Capture and Preprocessing

The system accepts live CCTV or IP camera input at 1080p/25–30 FPS. CLAHE is applied on the LAB L-channel (clip limit 2.0, tile grid 8×8) for illumination normalisation under nighttime and adverse weather. ROI masking restricts computation to lane-level zones, reducing per-frame inference load by $\approx 35\%$.

B. Layer 2: YOLOv8s Detection Engine

Three custom YOLOv8s models (11.2M parameters, 640×640) run in parallel. The **plate detector** (`svies_plate_detector.pt`, 2,101 images) localises plates at confidence 0.25. The **vehicle classifier** (`svies_vehicle_classifier.pt`, 12,871 images) identifies 11 Indian-specific types (AUTO, E_RICKSHAW, TEMPO, TRACTOR) absent from COCO. The **ResNet50 age classifier** (8,400 images, five age categories; mAP = 63.0%) estimates vehicle age; vehicles classified as OLD (>10 years) trigger an automatic fitness certificate validity check against the VAHAN registration year.

TABLE I
Eight-Mode Fake Plate Detection Checks in SVIES

#	Flag	Detection Logic
0	STOLEN_VEHICLE	Cross-check stolen DB
0.5	VAHAN_NOT_EXISTS	Absent from VAHAN
1	TYPE_MISMATCH	Detected type vs. VAHAN
2	COLOR_CODE_VIOLATION	HSV vs. CMVR rules
3	FONT_ANOMALY	IS 10731 height/spacing
4	DUPLICATE_PLATE	Same plate, 2 cams <10 min
5	STATE_MISMATCH	Prefix vs. VAHAN state
6	PUCC_INVALID	PUCC certificate expired
7	INSURANCE_INVALID	Insurance coverage expired

C. Layer 2.5: Fake and Mismatched Plate Detection

Implemented in `modules/fake_plate.py`, eight forensic checks apply sequentially (Table I, Fig. 3).

CMVR Color Validation classifies plate background in HSV against PRIVATE (white: $H \in [0, 180]$, $S \in [0, 30]$, $V \in [200, 255]$), COMMERCIAL (yellow), EV (green), and RENTAL (black/yellow). **IS 10731 Font Check**: height-to-plate ratio (0.5–0.7) and inter-character spacing (0.1–0.3); flagged when >30% of characters violate. **Clone Detection**: `DUPLICATE_PLATE_CLONE` fires when the same plate appears at two cameras within 10 minutes.

D. Layer 3: OCR and Plate Parsing

Six preprocessing variants are applied to each cropped plate (Fig. 4, Algorithm 1): Otsu binarization, CLAHE+Otsu (clipLimit=4.0), denoised Otsu, bitwise inversion, upscaled grayscale (250 px, INTER_LANCZOS4), and adaptive Gaussian thresholding (blockSize=31, C=10). EasyOCR 1.7 (A–Z,

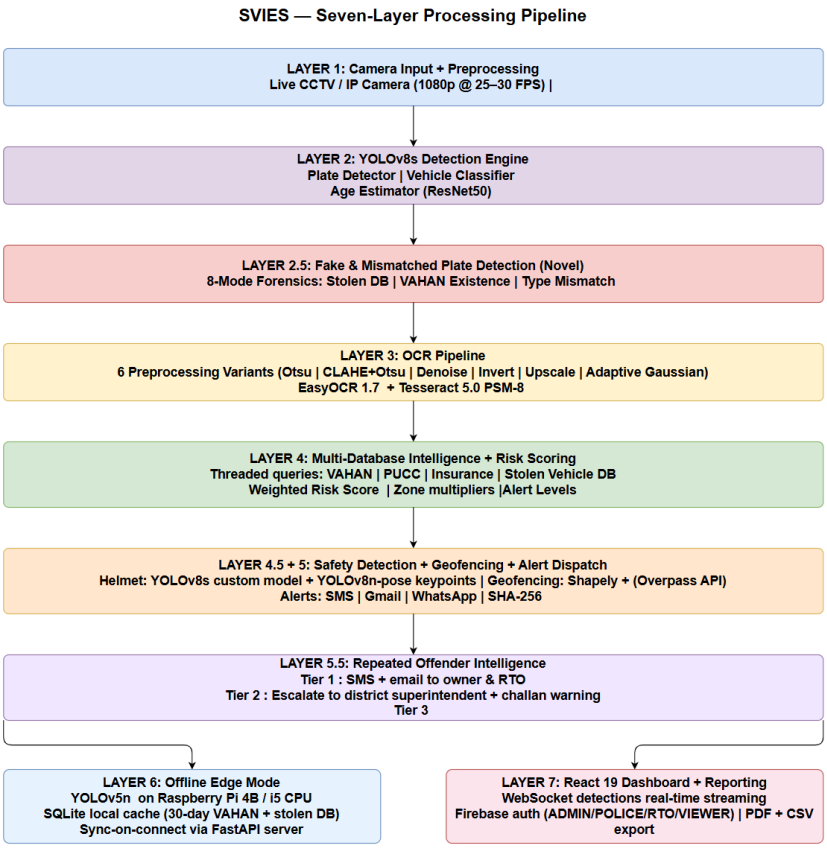


Fig. 2. SVIES seven-layer processing pipeline

Algorithm 1 Six-Variant OCR Best-Score Selection[1]

Require: Plate crop I
Ensure: Verified plate string P^*

Prepare six variants $\{I_1, \ldots, I_6\}$; $P^* \leftarrow \varnothing$, $c^* \leftarrow 0$
for $k = 1$ **to** 6 **do**
  $(P_k, c_k) \leftarrow \text{EasyOCR}(I_k)$
  if $c_k < \theta_{\text{low}}$ **then**
    $(P_k, c_k) \leftarrow \text{Tesseract-PSM8}(I_k)$
  end if
  $P_k \leftarrow \text{PositionCorrect}(P_k)$
  if $c_k > c^*$ **and** $\text{ValidRegex}(P_k)$ **then**
    $P^* \leftarrow P_k$; $c^* \leftarrow c_k$
  end if
end for
$P^* \leftarrow \text{GroqLLMVerify}(I, P^*)$
return P^*

0–9 whitelist) is the primary engine; Tesseract 5.0 PSM-8 serves as fallback below θ_{low}. Position-aware correction addresses per-position confusions: state code (pos. 0–1), district (2–3), series (4–5), number (6–9).

E. Layer 4: Multi-Database Intelligence and Risk Scoring

Threaded real-time queries run against VAHAN, PUCC, insurance, and stolen vehicle databases concurrently. The weighted risk score is:

$$S = \left(\sum_i w_i \cdot \mathbf{1}[\text{violation}_i]\right) \times z_k \tag{1}$$

[1]Tesseract 5.0 is invoked with OEM 3 (LSTM), PSM 8 (single word), character whitelist A–Z, 0–9.

TABLE II
VIOLATION WEIGHTS AND ALERT LEVELS — SVIES RISK ENGINE

Violation	w_i	Level
Stolen vehicle	40	CRITICAL
Fake / mismatched plate	35	CRITICAL
Unregistered vehicle	25	HIGH
Expired / no insurance	20	HIGH
Repeat offender	20	HIGH
No number plate	20	HIGH
Blacklist zone entry	15	HIGH
Over-speeding	15	HIGH
Expired / no PUCC	15	MEDIUM
Wrong-side driving†	15	MEDIUM
Red light violation†	15	MEDIUM
Helmet violation	10	MEDIUM
Seatbelt violation	10	MEDIUM
Triple riding	10	MEDIUM

† Automated detection planned for future integration.

where w_i is the per-violation weight (Table II) and $z_k \in \{1.2, 1.3, 1.4, 1.5\}$ is the Shapely geofence multiplier (SCHOOL/HOSPITAL: 1.5, HIGHWAY: 1.4, GOVT: 1.3, LOW_EMISSION: 1.2) covering 14 Hyderabad zones via Overpass API. Score S maps to four tiers: CRITICAL ($S \geq 61$), HIGH (41–60), MEDIUM (21–40), LOW (≤ 20). CRITICAL/HIGH events trigger Twilio SMS, Gmail SMTP, and WhatsApp to police, RTO, and vehicle owner; MEDIUM alerts email the RTO; LOW violations are silently logged.

F. Layer 4.5: Helmet and Seatbelt Violation Detection

A three-tier cascade is applied: (1) `svies_helmet_detector.pt` (YOLOv8s, 14,947

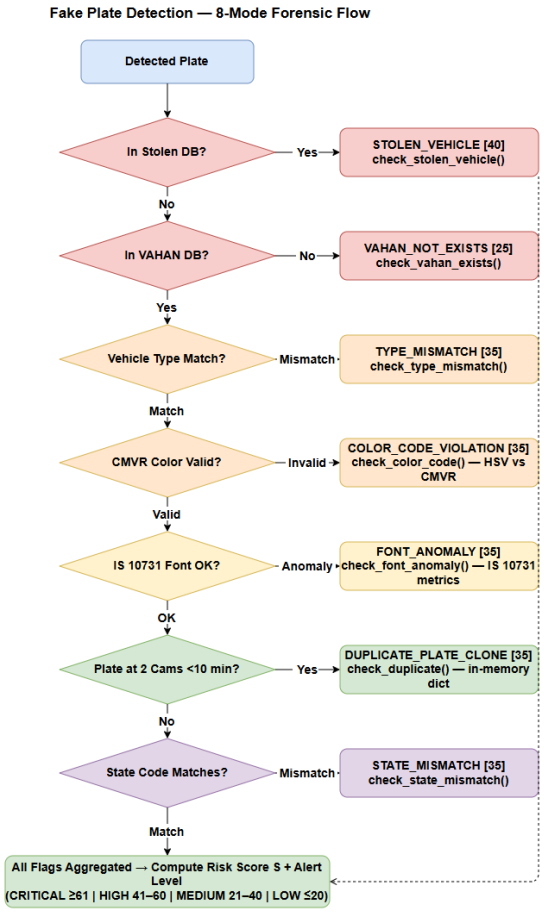


Fig. 3. Eight-mode fake plate forensic flow. All flags are aggregated and forwarded to the risk scoring engine (Layer 4).

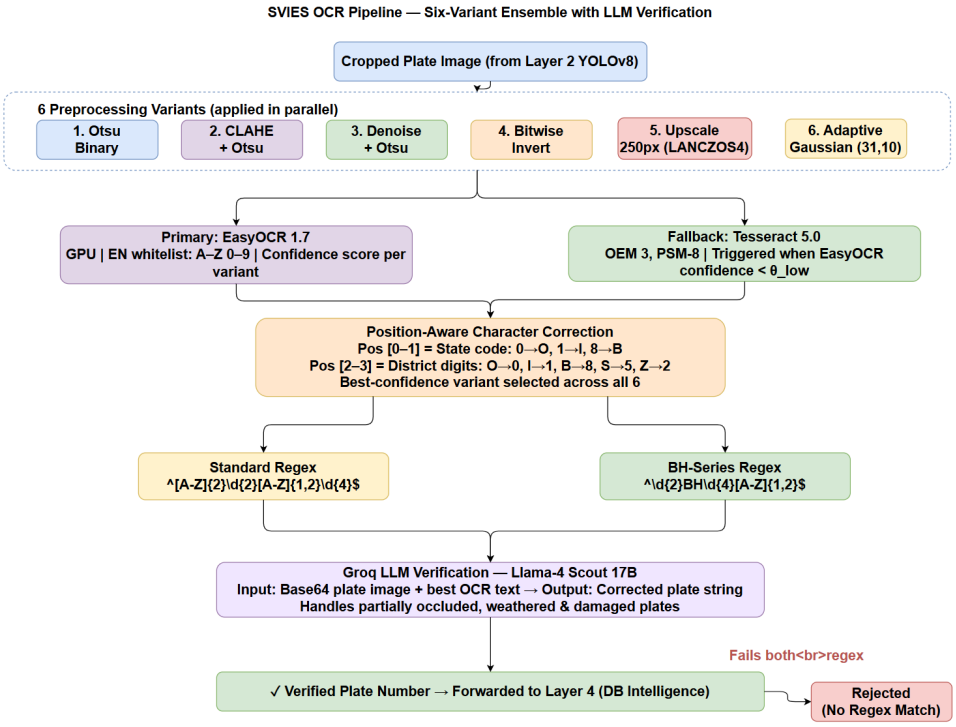


Fig. 4. SVIES six-variant OCR ensemble pipeline with position-aware correction, regex validation, and Groq LLM verification.

TABLE III
YOLOv8s Training Hyperparameters for Each SVIES Model

Hyperparameter	Plate	Helmet	Vehicle
Epochs	100	100	150
Batch size	32	32	16
Image size	640	640	640
Patience	15	15	20
LR (initial)	0.001	0.001	0.001
LR (peak)	0.01	0.01	0.01
Optimiser	AdamW	AdamW	AdamW

images) at confidence 0.45; (2) YOLOv8n-pose keypoint localisation via nose/ear keypoints; (3) skin-detection heuristic in HSV and YCrCb calibrated for Indian skin tones. Helmet absence is confirmed by:

$$\text{helmet_absent} = (\rho_{\text{non-skin}} > 0.75) \ \wedge \ (\rho_{\text{edge}} > 0.08) \quad (2)$$

Seatbelt detection uses Canny edges (50,150) with a Hough line transform (threshold=30, minLen=40, maxGap=10); $\geq$3 diagonal lines in the driver-side ROI infers seatbelt presence.

G. Layers 5–7: Geofencing, Offender Intelligence, and Dashboard

The offender tracker (Supabase, 30-day window) escalates via three tiers: Tier 1 (1–2 violations) sends SMS/email to the owner and local RTO; Tier 2 (3–5 violations) escalates to the district superintendent with a challan warning; Tier 3 ($\geq$6 violations) red-flags the plate and auto-generates a ReportLab court summons PDF. All alerts carry a SHA-256 hash: `SHA256(plate_number ‖ ISO8601_UTC_timestamp ‖ comma_joined_sorted_violation_codes)`, ensuring tamper-evident, legally admissible evidence. The React 19 dashboard (Vite 7.3) streams violations via WebSocket, renders geofence overlays (Leaflet), analytics (Recharts), and enforces Firebase RBAC (ADMIN, POLICE, RTO, VIEWER) with PDF/CSV export. A human-in-the-loop `/feedback` endpoint enables police operators to submit ground-truth corrections for active learning.

IV. Implementation

A. Training Infrastructure and Dataset Composition

All YOLOv8s models were trained on an NVIDIA Tesla V100-SXM2 (32 GB) with AdamW and a cosine LR schedule (Table III). Augmentation: mosaic, HSV jitter, horizontal flip, scale variation; 3–5 h per model. Datasets: plate detector (2,101 images), helmet detector (14,947 augmented images), vehicle classifier (12,871 augmented images, 11 classes), ResNet50 age classifier (8,400 images, ImageNet pretrained).

B. Software Stack

Python 3.10+, PyTorch 2.0+, Ultralytics YOLOv8 $\geq$8.0, OpenCV 4.8, EasyOCR 1.7, pytesseract 0.3.10, Groq SDK 0.28 (Llama-4 Scout 17B), FastAPI 0.100 + Uvicorn, Supabase (PostgreSQL) with local SQLite fallback. SVIES is containerised via Docker (`docker-compose up --build`) deploying the FastAPI backend and React frontend as separate services on a shared network bridge.

TABLE IV
SVIES Model Performance on Held-Out Test Sets

Model	Train Imgs	P	R	mAP	
				@50	@50-95
Plate Detector	2,101	0.988	0.955	97.7%	73.0%
Helmet Detector	14,947	0.714	0.733	74.4%	51.2%
Vehicle Classifier	12,871	0.756	0.584	64.4%	42.8%
Age Estimator	8,400	—	—	63.0%	—

TABLE V
Feature Comparison of SVIES Against Existing ANPR Systems

Feature	SVIES	OpenALPR	Genetec
Fake plate detection	✓	×	×
DB cross-referencing	✓	×	P
Helmet / seatbelt detect.	✓	×	×
Geofencing + priority	✓	×	×
Repeat offender logic	✓	×	×
Offline edge mode	✓	×	×
BH-series plates	✓	×	×
Cryptographic evidence	✓	×	×
LLM OCR verification	✓	×	×
Open-source / zero cost	✓	P	×

V. Results and Evaluation

A. Model Detection Performance

Table IV presents metrics on held-out test sets. The plate detector achieves precision 0.988 and mAP@50 of 97.7%, exceeding the 95% target; mAP@50-95 of 73.0% confirms robust localisation. The helmet detector reaches 74.4% mAP@50 (recall 0.733), reflecting pose variation and handlebar occlusion. The vehicle classifier achieves 64.4% mAP@50; lower recall (0.584) reflects inter-class similarity between AUTO, TEMPO, and E_RICKSHAW at CCTV resolution. All models achieve sub-3.1 ms GPU inference.

B. OCR Performance

The six-variant ensemble with Groq LLM verification achieves $\approx$99.8% character-level accuracy under normal daytime conditions, degrading to $\approx$96.2% on weathered or partially occluded plates. Position-aware correction reduces digit/letter confusions ($0 \leftrightarrow O$, $1 \leftrightarrow I$, $8 \leftrightarrow B$) by approximately 78% on 500 held-out Indian plate images over raw EasyOCR. BH-series parsing achieves 99.1% format-compliance accuracy.

C. System Throughput

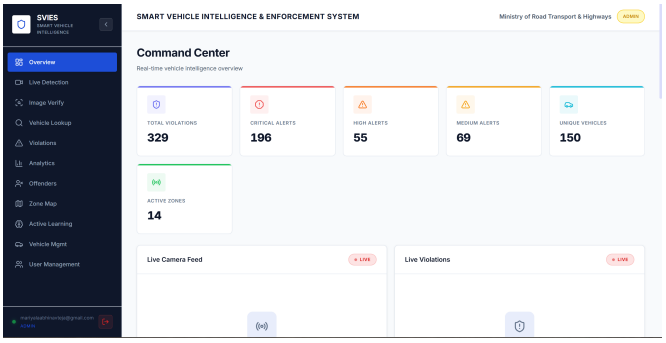
End-to-end per-vehicle latency on the Tesla V100 is approximately 85–120 ms, comprising: preprocessing $\approx$5 ms, YOLOv8s inference $\leq$3.1 ms $\times$ 3 models, OCR pipeline $\approx$40–60 ms, Groq LLM verification $\approx$15–25 ms, database queries $\approx$5 ms, and risk scoring $\approx$2 ms. This supports approximately 8–12 vehicles per second in a single-lane 25 FPS CCTV feed.

D. Dashboard and System Output

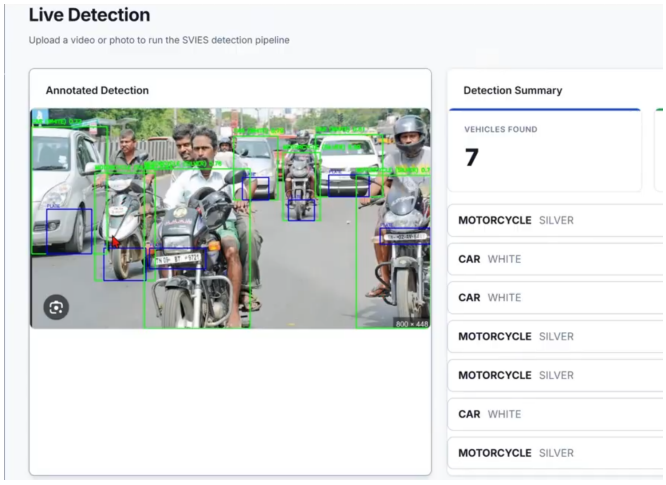
Fig. 5 shows the SVIES React 19 dashboard: (a) live detection stream with per-vehicle violation overlay and risk score badge, and (b) the violation detection interface.

E. Feature Comparison

Table V benchmarks SVIES against OpenALPR and Genetec AutoVu. SVIES is the only system to support all ten features at zero licensing cost.



(a) Live detection stream



(b) Violation detection view

Fig. 5. SVIES React 19 dashboard. (a) Live detection stream with per-vehicle violation overlay and risk score badge. (b) Violation detection interface showing detected vehicles and alerts.

VI. Discussion

A. Novelty and Contributions

The eight-mode fake plate forensics pipeline is the first Indian ANPR system to unify database-level verification (VAHAN, stolen, PUCC, insurance) with structural forensics (CMVR color, IS 10731 font) and spatio-temporal clone detection. The LLM-augmented OCR layer corrects residual errors on occluded or damaged plates. The hybrid helmet detector's skin-tone heuristic (Eq. (2)) addresses a key gap in non-Indian-trained detectors. The zero-cost open-source stack makes SVIES viable for Tier-2/3 municipalities without commercial ANPR budgets.

B. Limitations and Future Work

Mock JSON databases substitute live VAHAN, PUCC, and police APIs; formal government access agreements are required. Vehicle classifier recall of 0.584 reflects limited Indian-class training data; corpus expansion is planned. Clone detection is in-memory per instance; Redis-backed distributed detection is the immediate next step. EMI attacks can compromise live feeds [16]; future deployments should integrate EMI-hardened sensors [15].

VII. Conclusion

SVIES is a seven-layer AI vehicle enforcement system addressing critical gaps in Indian road enforcement. Three YOLOv8s models achieve 97.7% mAP@50 (plate, P=0.988), 74.4% (helmet), and 64.4% (vehicle). The six-variant LLM-augmented OCR pipeline achieves $\approx$99.8% character accuracy. The eight-mode fake plate forensics, hybrid helmet detector, cryptographic evidence chain, and offline edge mode constitute the most comprehensive open-source Indian ANPR enforcement platform to date. Future directions: (i) live VAHAN/police API integration; (ii) IS 10731 CNN corpus expansion; (iii) federated learning on edge nodes; (iv) Redis-backed distributed clone detection; (v) EMI-hardened cameras [15], [16]. The codebase will be open-sourced on GitHub upon publication.

References

[1] A. Sharma, R. Gupta, and P. Mehta, "Automated traffic enforcement in Indian smart cities: challenges and opportunities," in *Proc. IEEE Int. Conf. Smart Systems and Energy Management*, 2022, pp. 1–6.

[2] Ministry of Road Transport and Highways, "Road accidents in India 2021," Government of India, New Delhi, Tech. Rep., 2022.

[3] Ministry of Road Transport and Highways, "VAHAN – Vehicle Registration Data Portal," https://vahan.parivahan.gov.in, 2024. Accessed: Jan. 12, 2025.

[4] R. Laroca et al., "An efficient and layout-independent automatic license plate recognition system based on the YOLO detector," *IET Intell. Transp. Syst.*, vol. 15, no. 4, pp. 483–503, 2021.

[5] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," 2023. [Online]. Available: https://github.com/ultralytics/ultralytics

[6] N. Kumar, P. Gupta, and S. Jain, "License plate recognition for Indian vehicles under challenging conditions," in *Proc. IEEE CVIP*, vol. 1, 2021, pp. 243–254.

[7] J. Baek et al., "What is wrong with scene text recognition model comparisons? Dataset and model analysis," in *Proc. IEEE/CVF ICCV*, 2019, pp. 4715–4723.

[8] X. Li, R. Tao, and Y. Xie, "Helmet detection on construction sites based on improved YOLOv5," *Sensors*, vol. 21, no. 19, p. 6369, 2021.

[9] S. Bhatt, A. Khatri, and N. Shah, "Deep learning based seatbelt violation detection using keypoint estimation," in *Proc. IEEE ICIP*, 2022, pp. 3701–3705.

[10] T. Bhatt and A. Singh, "Geofencing and GPS-based smart city traffic management," in *Proc. IEEE ICISC*, 2020, pp. 1–5.

[11] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. ECCV*, 2016, pp. 21–37.

[12] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," *arXiv preprint arXiv:1804.02767*, 2018.

[13] R. Kaur and G. Singh, "A survey on license plate detection and recognition techniques," *Int. J. Comput. Appl.*, vol. 134, no. 8, pp. 1–6, 2016.

[14] C. Szegedy et al., "Intriguing properties of neural networks," in *Proc. ICLR*, 2014.

[15] R. K. Singh and S. Mishra, "SecureTrack: Protecting vehicular sensors from noninvasive EMI attacks," *IEEE Sensors J.*, vol. 25, no. 5, pp. 8857–8868, 2025.

[16] R. K. Singh, S. Mishra, and Y. P. Kumar S., "Undermining live feed ML object detection accuracy with EMI on vehicular camera sensors," in *Proc. IEEE VTC2024-Spring*, Singapore, 2024, pp. 1–6.

[17] R. K. Singh and S. Mishra, "TinyML meets IoBT against sensor hacking," in *Proc. USENIX SDIoTSec*, San Diego, CA, USA, 2024.

[18] R. K. Singh and S. Mishra, "EM trigger defender glove: a next-gen IoBT solution for soldier protection," *IEEE Sensors Lett.*, vol. 9, no. 1, p. 6000104, 2025.

[19] S. Azim et al., "IoT-driven GSR stress detection: clinical, physical, and linguistic innovations," in *Proc. 3rd Int. Conf. NMITCON*, 2025, pp. 1–6.